

ARYAN SURYA S

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PROFESSIONAL SUMMARY

Computer Science Engineering student at Dayananda Sagar University with hands-on full-stack development experience across **React**, **Node.js**, and **Python**, and a consistent record of self-directed projects spanning **3D web experiences**, **computer-vision applications**, and **data visualization tools**.

Oracle Cloud Infrastructure and **Google Generative AI** certified, with active DSA practice and a growing interest in applying **LLMs** to real products. Comfortable owning a project end to end — architecture, implementation, and deployment.

EDUCATIONS

Dayananda Sagar University | Bengaluru

2024 - 2028

B.Tech in Computer Science Engineering

- **Pre-University (PCMB)**, State Board — **93%**
- **10th Standard** — **96.96%**

PROJECTS

Parallax Earth — Cinematic Scroll Experience | Github

- Engineered a full-screen, scroll-driven cinematic web experience spanning **13.8 billion years** of cosmic history, mapping each scroll position to a distinct era with its own color palette, typography, and **Three.js** particle system.
- Built scroll-synchronized animation sequences with **GSAP** and **ScrollTrigger**, coordinating particle systems, starfield rendering, and a **Big Bang** flash effect across smooth era transitions.
- Integrated **Firebase Firestore** to power a live visitor counter, era-based voting, and a leaderboard, with graceful feature degradation when Firebase credentials are absent.
- Implemented era-based ambient audio with **Howler.js**, including automatic crossfading between soundtracks as the user scrolls between time periods.

Technologies / Tools Used : React, Vite, Three.js, GSAP, Firebase Firestore, Tailwind CSS, Framer Motion, Howler.js

Hand Frame FX — Gesture-Controlled Visual Effects App

- Developed a real-time computer-vision application that tracks both hands via **MediaPipe** and uses the index and thumb fingertips to frame a live quadrilateral, compositing stylized visual effects only inside that region of the camera feed.
- Implemented convex-hull mask generation (**OpenCV**) to render one of four custom visual effects (comic, paper, grid, pixel-glass) in real time over the webcam stream.
- Applied exponential moving average smoothing to landmark coordinates to eliminate jitter, with dropout tolerance for brief tracking losses.
- Designed a hands-free UX where bringing both hands together automatically cycles effects, alongside keyboard controls for switching, live recording, and a debug overlay.

Technologies / Tools Used : Python, OpenCV, MediaPipe Hands, NumPy

Hand Frame Addition — Real-Time Gesture Calculator

- Built a real-time finger-counting calculator that detects up to two hands using **MediaPipe's** 21-point hand landmark model and computes a live sum from the fingers raised on each hand.
- Designed position-based logic mapping the left hand to the first operand and the right hand to the second, with a live on-screen HUD showing the running addition equation.
- Implemented handedness-aware geometric rules to detect finger extension — vertical tip-to-knuckle comparison for the four fingers and horizontal comparison for the thumb — to reliably distinguish raised vs. curled fingers.

Technologies / Tools Used : Python, OpenCV, MediaPipe Hand Landmarker, NumPy

SKILLS

Databases :	MongoDB, MySQL, PostgreSQL
Frameworks & Libraries :	React.js, HTML5, CSS3, Tailwind CSS, Node.js, Express.js, Django
Programming Languages :	JavaScript (ES6+), Python, Java, C++, C, Rust
Soft Skills :	Data Structures & Algorithms, Object-Oriented Programming, Problem Solving, Debugging, Version Control
Tools & Platforms :	Git, GitHub, VS Code, Figma, Power BI, Oracle Cloud Infrastructure

AWARDS & ACHIEVEMENTS

- **Google Generative AI Fundamentals** — Large Language Models and AI integration
- **NPTEL** — Programming in Java (Elite Distinction)

CERTIFICATIONS